Govardhana Kondapaturi

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SUMMARY

Physical Design Engineer with 3+ years of ASIC implementation experience across AMD and Astera Labs, currently on 7 nm high-speed connectivity silicon. Hands-on floorplanning, power planning, placement, CTS, routing, and timing closure across PVT corners in Cadence Innovus with Synopsys PrimeTime MCMM signoff; SpyGlass/JasperGold CDC/RDC, Calibre DRC/LVS/ANTENNA, RedHawk-SC IR/EM. Drove a 180K-gate three-chiplet SoC from RTL to GDSII on Nangate 45 nm at 513 MHz and submitted an all-digital PLL to the SkyWater SKY130 shuttle on September 7, 2026. M.S. ECE, California State University, Fresno.

TECHNICAL SKILLS

- **Physical Design & Implementation:** Cadence Innovus, floorplanning, macro placement, power/clock distribution, CTS, routing, congestion analysis, timing closure, ECOs, multi-voltage design (UPF, clock gating, multi-Vt, isolation, level shifters)
- **Static Timing & Physical Verification:** Synopsys PrimeTime, MCMM setup/hold analysis, SDC constraints, signoff STA, RedHawk-SC IR-drop and EM, Calibre DRC/LVS/PEX/ANTENNA/ERC signoff, gate-level simulation with SDF annotation
- **RTL, Simulation, Synthesis & Formal:** Verilog, SystemVerilog, Cadence Xcelium simulation, Synopsys Design Compiler, DFT-aware synthesis, SpyGlass CDC/RDC and lint, JasperGold, Conformal and Formality equivalence checking
- **Programming & Automation:** Python, TCL, Perl, C/C++, STA report parsing, ECO/regression automation, LLM-augmented tooling

PROFESSIONAL EXPERIENCE

Physical Design Engineer | Astera Labs
San Jose, CA, USA | Aug 2025 – Present

- Worked on 7 nm ASIC physical design in Cadence Innovus: floorplanning, power planning, placement, CTS, and routing.
- Used Synopsys PrimeTime to analyze timing reports and resolve setup and hold violations across PVT corners.
- Investigated placement congestion and routing issues affecting critical timing paths in high-speed connectivity designs.
- Made cell-sizing, buffering, and ECO changes to address timing violations and improve timing margins.
- Checked clock-tree skew, insertion delay, and transition violations during timing closure.
- Compared pre-route and post-route timing reports to identify delays caused by parasitic effects.
- Reviewed IR-drop, electromigration (EM), DRC, LVS, and antenna-check reports during physical signoff.
- Used TCL and Python scripts to analyze timing reports and automate routine implementation checks.

Silicon Design Engineer | AMD
Bengaluru, Karnataka, India | Jun 2022 – Jul 2024

- Performed ASIC physical implementation tasks involving floorplan review, placement, and routing analysis.
- Examined placement reports to identify congestion hotspots and areas requiring further implementation changes.
- Reviewed clock-tree reports and investigated skew, latency, and transition issues.
- Analyzed static timing reports to identify critical paths and determine the cause of timing violations.
- Reviewed extracted parasitics to understand their effect on signal delays after routing.
- Checked DRC, LVS, and antenna reports and worked on resolving physical verification violations.
- Wrote TCL scripts to extract timing metrics, organize implementation reports, and simplify recurring design checks.

ACADEMIC PROJECTS

All-Digital PLL Tapeout on SkyWater SKY130 (TinyTapeout SKY26c Shuttle) **Fresno, USA**
Sole Designer (Academic) | Aug 2026 - Sept 2026, Silicon Submitted on Sept 7, 2026

- Designed an all-digital PLL (TDC, PI loop filter, ring-oscillator DCO, programmable divider) in synthesizable Verilog.
- Hardened in OpenLane/OpenROAD on SKY130: 40 MHz, sub-10% IR drop, clean DRC/LVS/ANTENNA, GDSII submitted.

Multi-Chiplet SoC with DVFS, CDC, and Hierarchical Power Management (RTL-to-GDSII) **Fresno, USA**
Team Lead (Academic) | Jan 2025 - May 2025

- Owned the full Innovus flow on a 180K-gate, 12 async-domain SoC to 513 MHz closure with MCMM PrimeTime signoff.
- Root-caused a 30 ps hold on die-to-die NPU-to-L2 AXI at 0.55 V; fixed by local skew rebalance, no setup regression.

EDUCATION

Master of Science in Electrical and Computer Engineering **Fresno, USA**
California State University | GPA: 3.3/4.0

Bachelor of Technology in Electronics and Communication Engineering **Vijayawada, India**
Vasireddy Venkatadri Institute of Technology | GPA: 3.56/4

PUBLICATIONS

- G. Kondapaturi and A. Stillmaker, "A UPF-Driven Multi-Voltage Three-Chiplet SoC with DVFS, carried from RTL to GDSII," submitted to IEEE LASCAS 2027.

CERTIFICATIONS

- **Synopsys:** PrimeTime, Fusion Compiler Design Implementation. **Cadence:** RTL-to-GDSII v7.0 (Advanced), SystemVerilog for Design & Verification.